



National Institute of Technology, Rourkela
Department of Computer Science and Engineering

B. Tech/M. Tech.(IS)/Ph.D./PhD(Ex.) End-Semester Examination (Autumn) 2023

Subject: **Foundations of Computer Security**

Subject Code: **CS 6301**

Full Marks: **50**

Duration: **3 Hours**

Answer **ALL** questions. *Figures at the right margin indicate marks.*

All parts of a question must be answered at one place.

Use of non-programmable calculator is allowed

1. (a) Explain with neat diagram Asymmetric crypto system. Consider a communication between Alice and Bob based on ElGamal crypto-system. Bob selects d , e_1 and p as 67, 2, and 107 respectively. What would be the public parameters? Alice wants to encrypt the letter 'B' and send it to Bob. What will be the cipher text? How Bob will decrypt the message? [4]
(b) Sweta wants to use RSA encryption scheme to send the message "dad is sad" to Rahul. Sweta selects a prime number p as 107 and find the value of $\Phi(n)$ as 11872. Calculate the cipher text, and also show how Rahul recovers the message. (*Hint: message only contains English alphabet [a-z] with a mapping to [0-25], space may be ignored*) [4]
(c) Define Euler's Totient Function. Find $\Phi(289)$, $\Phi(1000)$. [2]
2. (a) With suitable block diagram, describe the key generation that takes place in each round of AES-256. [4]
(b) Encrypt the plain text block 01001 using the super increasing sequence $\{3, 4, 10, 20, 42\}$ with modulus $m = 90$ and multiplier $a=17$. Consider the permutation table $[5, 2, 4, 3, 1]$. What cipher text Bob will receive and how he gets the plain text? [4]
(c) Is it possible to get the value of p and q for a given $n (= p \times q)$ and $\Phi(n)$? If yes then how can we find the factors p and q . [2]
3. (a) Solve the following simultaneous congruences
 $x \equiv 6 \pmod{11}$, $x \equiv 13 \pmod{16}$, $x \equiv 9 \pmod{21}$, $x \equiv 19 \pmod{25}$. [4]
(b) The propagating cipher block chaining (PCBC) mode is a variant of CBC in which both the previous plain text block and the previous cipher text block are exclusive-ored with the current plain text block before encryption. Draw a diagram that shows the encryption and decryption [4]
(c) What is Quadratic Residue (QR) and Non-residue (QNR)? How can we check to see if an integer is a QR modulo p ? [2]
4. (a) Find all solutions of $x^2 \equiv 1 \pmod{144}$. [4]
(b) What is the difference between substitution cipher and transposition cipher? Explain any two method in each category of cipher. Construct the key matrix using the keyword "DOMESTIC" and encrypt the message "India tour to South Africa" using play fair cipher. [4]
(c) What is one-way trapdoor function? Explain with an example. [2]
5. (a) What is Cipher text stealing? Explain its use in ECB and CBC mode. [5]
(b) Find the results of the following [5]
 $145^{43} \pmod{49}$
 $70^{-1} \pmod{101}$
 $27^{-1} \pmod{41}$
 $84^{23} \pmod{35}$
 $456^{17} \pmod{17}$.

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